Project description

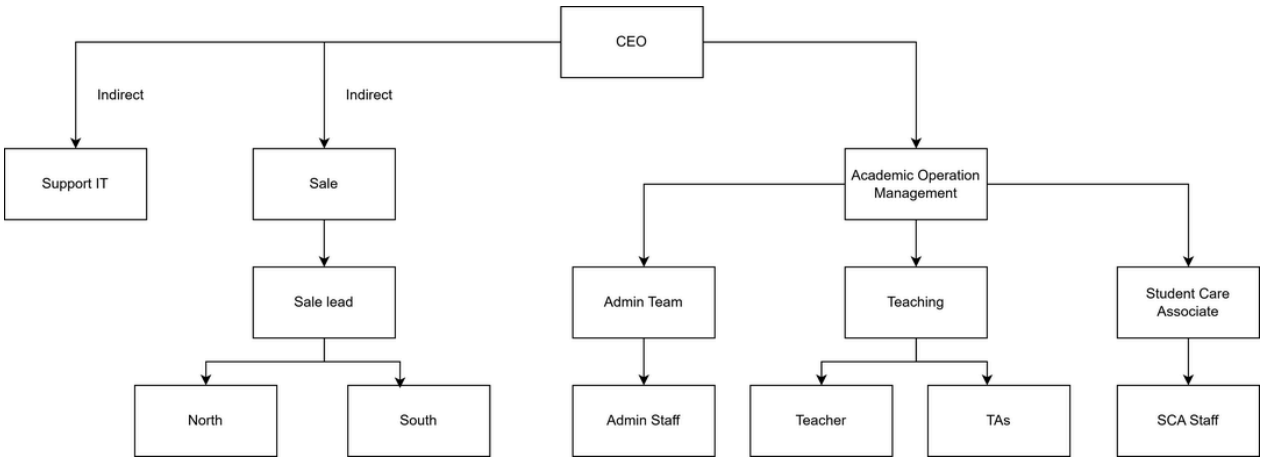
1 Project title

Integrated Learning Center Management System (ILCMS)

2 Customer description

The customer of this project is an online learning center that has been operating for approximately five years, since September 2021. The center currently serves around 300–500 students per year and operates about 15 active classes at any given time. Its teaching staff includes 45 teachers and 15–20 teaching assistants. The center provides only online learning, with all teaching, learning, and assessment activities conducted through digital platforms.

To manage daily operations, the center relies on multiple separate systems, including ECO, CMS, LMS, reporting tools, OTP2, ClassIn/ClassSeen, and Zoom. Academic administrators are responsible for most operational tasks such as creating student accounts, setting up classes, assigning teachers, managing schedules, activating online sessions, entering exam scores, and producing academic reports for parents. Due to the fragmented nature of these systems, administrators must frequently switch between platforms and manually verify data synchronization, which leads to inefficiencies and data inconsistency.



3 Business Process

3.1 Functional Areas

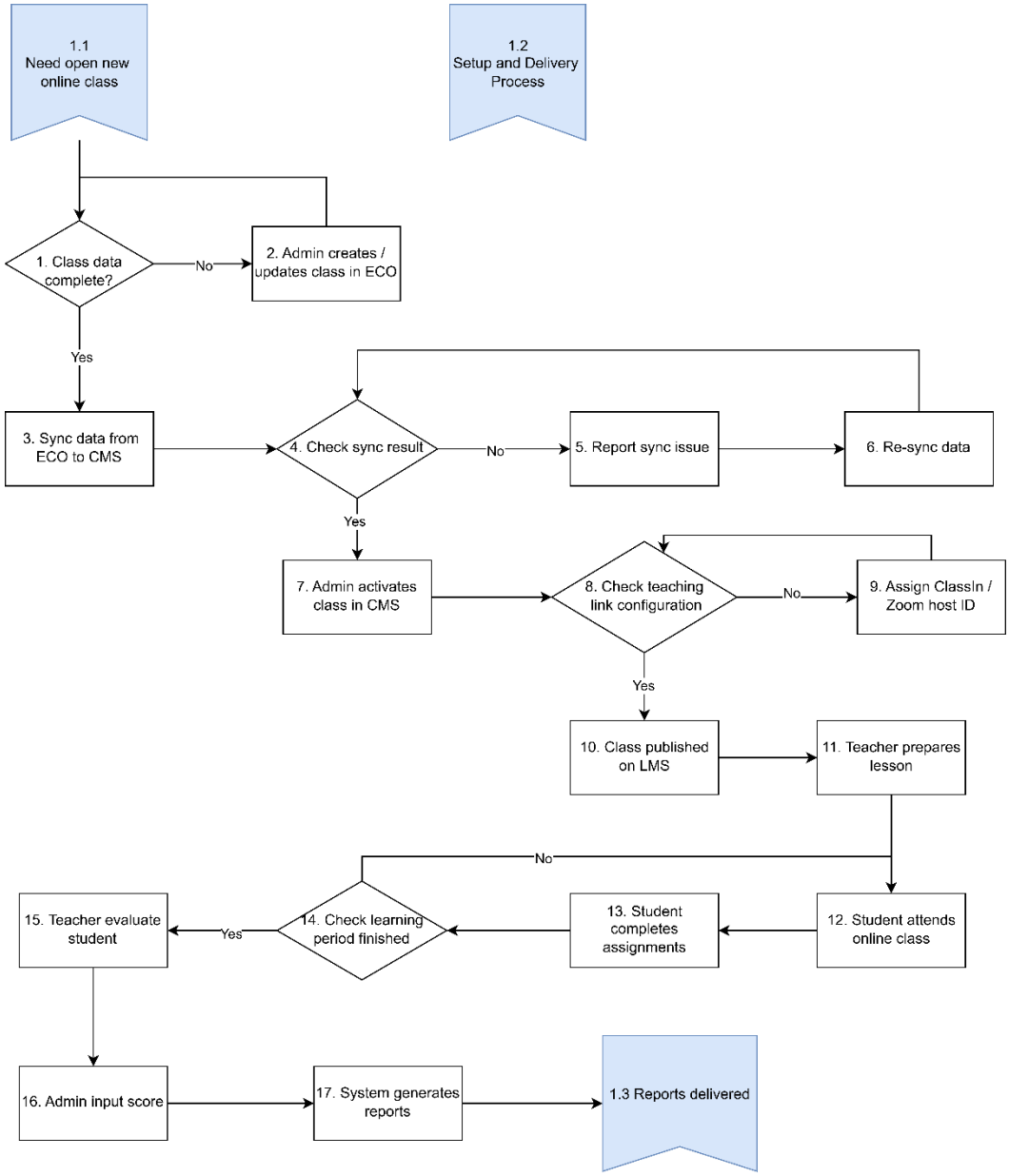
The business functions as an online learning hub, organized around a number of crucial functional domains that work together to provide top-notch instruction and student assistance:

- **Academic operations management:** Scheduling, academic reporting, instructor assignments, and class preparation and execution.
- **Teaching department:** Made up of instructors and teaching assistants who are in charge of delivering lessons, involving students, and conducting assessments.
- **Student Care Team:** Responds to questions, facilitates communication, and offers assistance to parents and students.
- **Administrative Team:** Manages student accounts, class creation, system configuration, and data entry across platforms.
- **Sales Department:** Customer relationship management, North and South regional outreach, and student registration.
- **IT Support:** Ensures platform integration, fixes system problems, and maintains technical infrastructure.

To carry out their duties, these departments rely on a variety of digital platforms (ECO, CMS, LMS, OTP2, ClassIn/ClassSeen, Zoom). However, the absence of connection across various platforms presents operational issues, requiring manual synchronization and cross-checking of data.

3.2 Core Cross-Functional Business Process: Online Class Setup and Delivery

The administrative team, academic operations, IT support, and teaching department all work together on this process. Below is a diagram and explanation of the workflow:



4 Project summary

The Integrated Learning Center Management System (ILCMS) aims to unify the fragmented operational processes of an online learning center into a single integrated platform. The project focuses on simplifying class setup, student and teacher management, online session activation,

assessment handling, and academic reporting. By reducing manual data synchronization and eliminating the need to switch between multiple systems, ILCMS will improve operational efficiency, reduce errors, and enhance transparency for administrators, teachers, students, and parents. The system is expected to deliver both economic benefits, through reduced administrative workload, and social benefits, by improving the learning experience and communication quality.

5 Significance and innovation

This project is undertaken to address critical operational inefficiencies caused by the use of multiple disconnected systems. The key innovation of ILCMS lies in its **integrated architecture**, which consolidates class management, learning management, online teaching, and reporting into a unified system. Instead of relying on manual synchronization and repeated verification, the proposed system provides real-time data consistency and centralized reporting. The project introduces improved workflow automation and role-based access, enabling administrators, teachers, and students to interact more efficiently with the system.

6 Approach

The project will follow an **Agile** development methodology, allowing continuous feedback from stakeholders. The system will be developed using a web-based architecture with modern frameworks (e.g., Java/Spring Boot or Node.js for backend, React for frontend) and a centralized relational database. APIs will be used to integrate online teaching tools. Emphasis will be placed on usability, data integrity, and scalability.

7 Expected outcomes

The expected outcome of this project is a fully integrated management system that significantly reduces administrative workload and data inconsistency. Academic administrators will be able to create and manage classes, students, and teachers through a single interface. Teachers will experience smoother access to class schedules, teaching materials, and assessment tools. Students will benefit from a more consistent learning experience with clear access to classes, assignments, and feedback. Overall, the system is expected to improve operational efficiency, data accuracy, and stakeholder satisfaction while supporting the long-term growth of the learning center.

8 Description of personnel

Name	Role	Responsibilities	Contributions
Nguyen Quoc Viet	Developer	Design backend & frontend; Develop core features; Integrate APIs; Optimize performance & security	Builds the technical foundation, ensuring the system is functional, secure and scalable
Vu Khanh Linh	Business Analyst	Gather requirements; Document workflows; Create user stories & diagrams; Validate business logic	Ensures the system reflects real-world needs and aligns with stakeholder expectations
Dinh Thi Hai	Project Manager	Define scope & timeline; Allocate resources; Manage communication; Monitor risks & progress	Provides strategic oversight, keeps project on tack and ensures time delivery
Nguyen Thi Hoang Yen	Tester	Design test cases; Perform functional & regression testing; Identify bugs; Validate user experience	Guarantees system reliability, usability and quality standards
Vu Lan Huong	Client Representative	Provide insights into workflows; Collaborate on requirements; Participate in testing; Act as liaison	Ensures the system meets operational needs and supports long-term growth

9 Feasibility Study

Even though the Integrated Learning Center Management System (ILCMS) hasn't been put into use yet, an initial feasibility study shows that it has a high chance of success in a number of areas.

9.1 Technical Feasibility

Widely used technologies like Java/Spring Boot or Node.js for the backend, React.js for the frontend, and relational databases like PostgreSQL/MySQL will be used in the development of the suggested system. It is technically possible to integrate existing platforms (Zoom, ClassIn, LMS) via APIs. Scalability and dependability can be guaranteed by containerization and cloud deployment. The technical viability of these technologies is high due to their maturity and the availability of qualified developers.

9.2 Operational Feasibility

Administrators, teachers, and students are already acclimated to digital platforms, which lessens the adoption hurdle. By combining disparate workflows into a single interface, the technology is designed to boost efficiency and eliminate errors. Stakeholders have expressed openness to adopt a more integrated solution, making operational feasibility promising.

9.3 Economic Feasibility

Long-term advantages include less administrative labor, cheaper maintenance expenses, and increased productivity, even if an initial investment in development and implementation will be necessary. The technology is projected to offer a good return on investment by streamlining processes and supporting scalability without commensurate increases in cost.

9.4 Social Feasibility

The system is projected to promote transparency, communication, and confidence among administrators, instructors, students, and parents. It will increase stakeholder satisfaction and improve the institution's reputation by enhancing the learning process and reporting accuracy.